

Earth-State Embeddings:

a self-supervised codec of the North Atlantic, its transport read-out,
and what a transformer forecasts from it

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Abstract

We compress the observed state of each North Atlantic grid cell — ocean reanalysis, Argo climatology, wind stress, sea-surface temperature — into a 32-dimensional embedding with a self-supervised per-pixel codec, and train a transformer to predict the embedding forward at five-day cadence. **Codec.** Read out by an attention head over the individual pixels of the 26.5°N section, the frozen embeddings predict RAPID overturning transport at $r = 0.67$ (two codec seeds, year-blocked k-fold); the same head fed raw section values reaches 0.66, so the representation adds nothing measurable for the current-state read-out. Wind stress accounts for most monthly skill (a wind-only ridge reaches 0.53); the 16 Argo levels move skill to the density-driven MOVE transport at 16°N ; neither codec capacity (0.92M to 40.7M) nor training length (50k to 1M steps) moves the probe. **Forecaster.** Under a training pool that excludes every window touching a held-out year, stage-2 heads from 7.6M to 400M parameters reach their best held-out one-step loss inside the first $\sim 2,000$ of 200,000 steps and worsen thereafter, the best level lying in 0.58–0.62 of the persistence error across a $53\times$ span in capacity. Rolled twelve months from held-out starts, the 206.7M head’s field anomaly correlation decays from 0.61 at five days to 0.41 at ten, ≈ 0.1 from two months on and -0.03 at a year; it beats raw persistence at every lead but the last and damped persistence at none. Its squared-error score is negative (mean MSSS -0.44 against climatology) because it emits anomalies at 78% amplitude at 10% correlation — an identity reproduces all 73 leads to 0.0135, and the same states rescaled would score $\approx +0.02$. Skill that survives past a few pentads lives in sea-surface temperature; 32 of 40 channels score nothing at any lead. A 200-mode linear inverse model fitted to the same field and rolled from the same starts is less correlated than the head at five and ten days and more correlated at every lead from fifteen days to a year (0.26 at 90 days, 0.18 at 365), with a squared-error score that stays positive throughout. Heads of 7.6M and 40.4M parameters rolled through the same battery have the same mean correlation as the 206.7M head (0.103, 0.104, 0.105); their squared-error scores differ (-0.18 , -0.26 , -0.44) only through an emitted amplitude that grows with width, and the smallest head does not decay through zero, holding ≈ 0.11 from two months to a year. Transport at 26.5°N shows no measurable skill at roughly nine effective starts. Every rolled number is a single seed. **Design.** From these constraints we specify the next codec and forecaster (§4). Both stages sample the same space–time hourglass — a learnable past cone as input, its point mirror as the forecast target — the codec over raw values at up to six pentads, the forecaster over the codec’s embeddings at log-spaced lags to two years. The forecaster is a residual on a contractive linear propagator initialised from the linear inverse model, trained on direct multi-lead targets under a decoupled Gaussian likelihood, and every choice is tied to the comparison on the dense field that would accept or reject it.

1 Data

The working tensor covers $0\text{--}70^\circ\text{N}$, $100^\circ\text{W}\text{--}20^\circ\text{E}$ on a 0.25° grid (281×481 cells, 86,698 ocean) at five-day cadence from 1982-01 to 2024-12 (3,142 bins), with 40 channels per cell: surface current speed, log mixed-layer depth and sea-surface height from the GLORYS reanalysis; temperature and salinity at 16 pressure levels from the Roemmich–Gilson Argo climatology (native 1° , monthly, from 2004); NCEP Reanalysis-1 wind stress and its within-bin variability (four channels); OISST sea-surface temperature. Channels absent before their record begins enter as missing tokens. Earlier codec generations used 1° monthly tensors with 12–25 channels, on the same window or globally; they are identified by channel count below. The transport truth is the RAPID-MOCHA overturning at 26.5°N (2004–), deseasonalised; MOVE at 16°N is a second target. Everything cited here — tensors, weights, result bundles — is public (Appendix A).

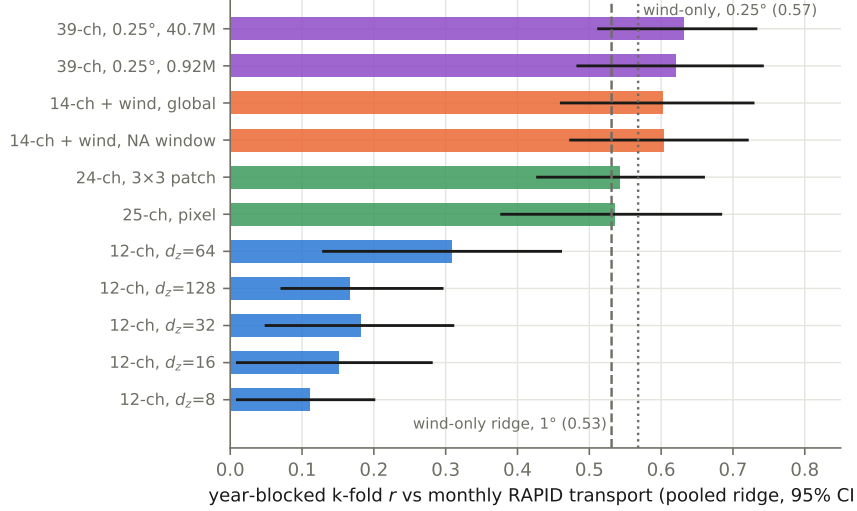


Figure 1: Pooled k-fold correlation with monthly deseasonalised RAPID transport by codec generation. Blue: the 12-channel width sweep; orange: with wind stress; green: 24–25 channels with 8 Argo levels; violet: the 0.25° tensor with 16 levels and SSH at two capacities.

2 The codec and its read-outs

Model. A masked autoencoder over one pixel’s channels at one time bin produces a d_z -dimensional embedding; a 3×3 -patch variant sees the pixel’s neighbours. The working codec on the 0.25° pentad tensor is 512-wide, 12 layers, decoder width 256, $d_z=32$, patch 1, 37.976M parameters, trained 200k steps at batch 512 with 2009, 2017 and 2023 held out of training and of the anomaly statistics. Embeddings are probed for transport with a year-blocked k-fold: a ridge on the section-mean embedding (*pooled*), or an attention head over the section’s individual pixels (*unpooled*). Field reconstruction is scored as the percentage improvement over persistence, one bin ahead.

What moves the transport read-out. Figure 1 is the pooled ranking across generations. Adding wind stress to a 12-channel tensor takes the pooled probe from 0.31 to 0.60; a wind-only ridge with no embedding reaches 0.531 on the 1° tensors and 0.568 on the 0.25° one, so most monthly skill at 26.5°N is Ekman physics. Training on the global ocean instead of the window costs the Atlantic read-out nothing (0.602 against 0.604). Bottleneck width matters little for reconstruction (saturating by $d_z=16$ – 32) and peaks at $d_z=64$ for the probe, turning over at 128 against ~ 220 truth months per fold. On the 0.25° tensor a 0.92M pilot reads 0.620 and the 40.7M codec 0.631 — $44\times$ the parameters buys $+0.011$, inside seed noise (± 0.05 on the pooled probe) — and a codec trained from 50k to one million steps is flat from the first probe to the last. Neither capacity nor training length is what stands between this codec and a better transport number.

Pooling was hiding the signal, and the codec is not what recovers it. The same frozen embeddings read three ways (Figure 2): an MLP on the section mean does no better than the ridge, so the ridge was never the limit as a function class; the attention head over the unpooled section beats every pooled probe and lifts the 3×3 -patch codec to 0.690 [0.570, 0.781] (RMSE 2.02 Sv against a 2.79 Sv target). Geostrophic transport is the east–west contrast across the section, and a section mean is the statistic in which that contrast cancels. The same head fed raw section anomalies instead of embeddings completes the attribution:

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3×3 neighbourhood	0.659 [0.55, 0.75]	0.690 / 0.654 (two codec seeds)

Unpooling the section is the whole of the improvement (pooled 0.54 $\rightarrow$ head 0.65+); the 3×3 receptive field adds $\approx +0.045$ and does so on raw data alone; pretraining adds $+0.013$ at two seeds,

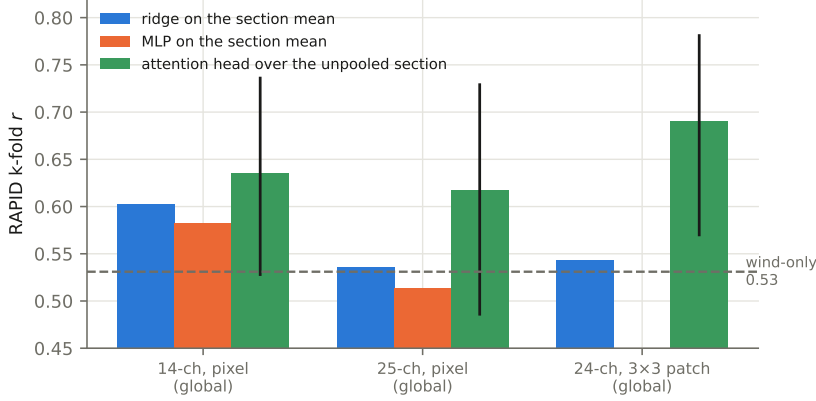


Figure 2: Same frozen embeddings, same folds, three read-outs. Head bars carry 95% intervals.

which is nothing against a head seed spread of 0.036. On the 0.25° monthly tensor the margin is $+0.034$ (0.662 against 0.628, one seed). At pentad cadence ($n=1,459$ transport bins) three unpooled read-outs — the frozen monthly codec (0.691), a codec trained on the pentad tensor itself (0.680), and raw fields with no codec (0.683) — land within 0.011 of one another, while the pooled ridge (0.660) sits below the pentad wind bar (0.670). For the current-state transport read-out, read-out design and not representation learning is what moved the number.

Where the deep channels go. Extending Argo from 4 to 16 levels did not raise the RAPID probe but roughly doubled skill at MOVE — a deep, density-driven transport — from 0.111 to 0.206 [0.044, 0.346], the first multi-target result whose interval excludes zero, with the 18-month low-passed correlation rising from 0.38 to 0.62. At 16°N the wind-only ridge is negative (-0.376). One task-blind representation serves a wind-dominated and a density-dominated target when the read-out respects the structure of the question.

Event anatomy. Out of fold, the winter 2009–10 collapse — the sharpest event in the RAPID record — is captured at 16% of its amplitude without wind channels, 50% with them, 59% by the 25-channel codec and 51% by the 40.7M 0.25° codec. Average correlation and event capture are different capabilities; the 25-channel codec has the best dip and the second worst monthly r .

Quantizing the embedding. A finite-scalar-quantized bottleneck (FSQ: each of $d_z=6$ dimensions rounded to 8, 8, 8, 5, 5 or 5 levels, one 2^{16} -way code per pixel and bin, a LayerNorm bound on the pre-quantization activation) trained from a random initialisation collapses on the pentad tensor: the encoder becomes input-independent within a few thousand steps (fitted level-spread $0.638 \rightarrow 0.005$ by step 7,500, twice), and an unbounded 8-level lattice on $d_z=32$ drifts to pre-quantization magnitudes $\sim 10^4$ against a lattice radius of 2, wearing its eight levels as a one-bit sign code. Switching the same lattice on at step 200k of a finished continuous $d_z=6$ codec and training 60k further steps keeps the activations input-dependent throughout (level-spread $0.97 \rightarrow 0.80$), at an 18% reconstruction tax (0.270 against 0.229) and a per-channel one-step error of 0.742 against 0.681 (persistence 0.843). The quantized codec reads RAPID at 0.588 [0.534, 0.641] against the continuous 0.579 [0.525, 0.627] through the unpooled head, one seed each — indistinguishable, and both below the raw 3×3 control at this width (0.693).

3 The forecaster

Model and protocol. A causal transformer over a window of $K=144$ consecutive embeddings at one pixel (two years) plus a 145-point spatial stencil on a spiral ring predicts the embedding one bin ahead, with input noise at 0.7 of the persistence scale, gradient clipping at 128, batch 256, learning rate 10^{-3} decaying exponentially (half-life 40k steps, warm-up 2k). The loss is dense over all 144 window positions, so the pool admits only windows that touch no held-out bin anywhere the forward pass can see (209,549,066 windows: $2,417$ end-times $\times 86,698$ pixels); it certifies this by brute force before training.

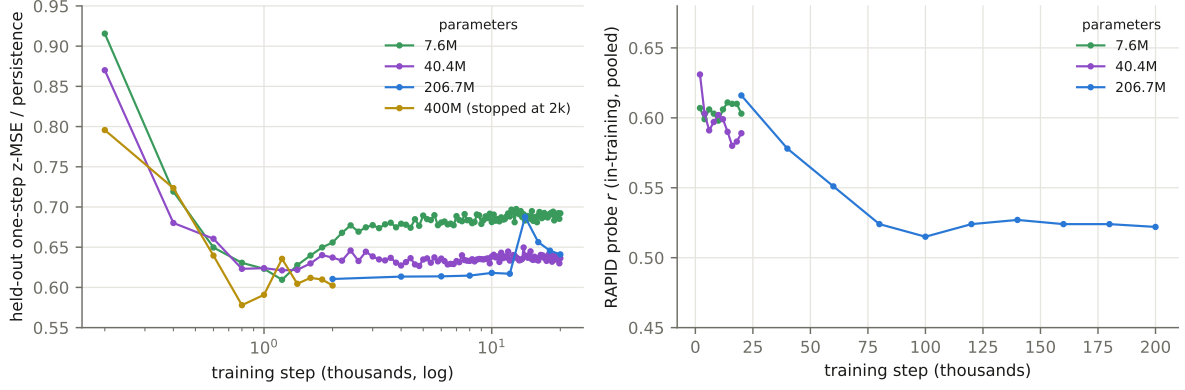


Figure 3: Left: held-out one-step ratio over the first 20,000 steps, by width. Right: the in-training RAPID probe over the full run.

params	best held-out ratio	at step	ratio at 20k	ratio at end	train ratio at end
7.598M	0.6095	1,200	0.692	0.692 (20k)	0.146
40.388M	0.6212	1,200	0.636	0.636 (20k)	0.086
206.659M	0.6105	$\leq 2,000$	0.641	0.686 (200k)	0.023
399.948M	0.5779	800	—	0.602 (2k)	0.171

Table 1: Held-out one-step ratio by width. The 206.659M run is recorded every 2,000 steps, so its minimum is at or before its first record; the others, recorded every 200, bottom out at 1,200.

The held-out one-step loss is the z -space MSE relative to persistence on held-out windows (denominator 21.44621). Widths: 256×8 (7.598M), 512×12 (40.388M), 1024×16 (206.659M), 1280×20 (399.948M); all on the frozen 37.976M codec and its embedding of the tensor.

A *roll* advances every ocean pixel one bin at a time from a true 144-bin context, feeding predictions back, for 73 steps (365 days), decodes through the codec and scores standardised anomalies per pixel and channel on three scopes: gate (600 random pixels plus the section), corridor (the fastest quarter of the window by current speed, dilated two cells, plus the section; 30,158 pixels), window (all pixels). Starts are three per held-out year, nine in all. Per lead: $\text{MSSS}_x = 1 - \text{MSE}/\text{MSE}_x$ against climatology (zero anomaly), raw persistence, and damped persistence (per-pixel AR(1) fitted on training years); anomaly correlation ACC; amplitude ratio a . The transport read-out regresses the rolled section states onto RAPID and reports r in three lead bands.

The held-out minimum is early at every width. Figure 3 and Table 1. All four widths reach their best held-out ratio inside 2,000 steps and get worse afterwards; the best levels span 0.578–0.621 across $53\times$ in parameters. At step 20,000 the 7.598M arm (0.692) is 0.051 worse than the 206.659M arm (0.641), and the ordering is not monotone in width (40.388M reads 0.636). Width buys how fast an arm reaches its minimum and how badly it then degrades, not the level of the minimum. The in-training transport probe separates the arms the other way: 7.598M holds 0.598–0.611 across ten probes, 40.388M drifts to 0.589, and 206.659M falls from 0.616 at 20k to 0.522 at 200k. The training loss keeps falling throughout (to a ratio of 0.023 at 200k for the 206.659M head, a $30\times$ train/held-out gap): the model memorises the training years long after it has learned anything that transfers.

The twelve-month roll. The 206.659M head at its 200k end state, rolled from the nine held-out starts (Table 2, Figure 4). Corridor anomaly correlation is 0.606 at five days, 0.412 at ten, 0.265 at fifteen, 0.20 at a month, near 0.1 from two months on, -0.03 at a year. The head beats raw persistence at 72 of 73 leads (mean $\text{MSSS}_{\text{pers}} + 0.204$) and damped persistence at none: $\text{MSSS}_{\text{damped}}$ runs from -0.011 at five days to -0.719 at a year (mean -0.461 ; gate -0.416 , window -0.420). A per-pixel AR(1) decay toward climatology, fitted on training years, is a better forecast of this field than the transformer at every horizon.

lead (days)	206.659M head				LIM, 200 modes			
	MSSS _{clim}	MSSS _{pers}	MSSS _{damped}	ACC	a	MSSS _{clim}	MSSS _{damped}	ACC
5	+0.365	+0.194	−0.011	0.606	0.692	+0.169	−0.323	0.406
10	+0.109	+0.252	−0.131	0.412	0.673	+0.135	−0.097	0.361
15	−0.172	+0.178	−0.354	0.265	0.761	+0.106	−0.034	0.324
30	−0.168	+0.252	−0.244	0.204	0.672	+0.069	+0.008	0.258
60	−0.320	+0.201	−0.342	0.107	0.690	+0.056	+0.040	0.227
90	−0.322	+0.211	−0.333	0.132	0.725	+0.075	+0.067	0.261
180	−0.354	+0.337	−0.357	0.153	0.778	+0.053	+0.051	0.223
270	−0.479	+0.227	−0.479	0.009	0.720	+0.034	+0.034	0.191
365	−0.719	−0.101	−0.719	−0.031	0.822	+0.019	+0.019	0.177
mean of 73 leads	−0.439	+0.204	−0.461	0.105	0.780	+0.054	+0.034	0.225

Table 2: The 206.659M head and the 200-mode linear inverse model, rolled from the same nine held-out starts, corridor scope. Head, gate and window scopes: mean MSSS_{clim} −0.395 and −0.401, mean ACC 0.131 and 0.121; LIM: +0.061 and +0.053, 0.253 and 0.233. The LIM’s amplitude ratio falls from 0.48 at five days to 0.06 at a year. Single seed for the head.

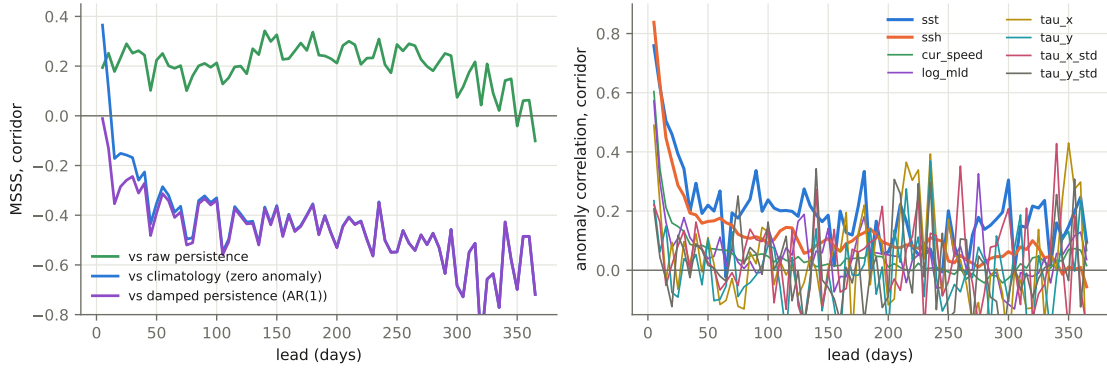


Figure 4: Left: the rolled head against its three references, corridor scope. Right: anomaly correlation against lead per scoreable channel; the 32 Argo channels have no scored samples at any lead.

The negative score is amplitude, not sign. For a forecast with correlation ACC and amplitude ratio a against a zero-anomaly reference, $\text{MSSS}_{\text{clim}} = 1 - (1 + a^2 - 2a \text{ACC})$ identically. From the head’s own per-lead ACC and a this reproduces the 73 measured values to a mean absolute error of 0.0135 (Figure 5). The head emits anomalies at $a \approx 0.78$ while its correlation is 0.1; rescaled to $a = \text{ACC}$ the same states would score ACC^2 per lead, a corridor mean of about +0.02. Whether a calibration fitted on training-year starts transfers to held-out starts is untested.

A linear inverse model. A linear inverse model fitted to the same standardised anomaly field — the eight scoreable channels over the window’s 86,698 ocean pixels, reduced by PCA to K modes, with the propagator $\mathbf{G}(\tau) = \mathbf{C}(\tau)\mathbf{C}(0)^{-1}$ estimated at τ of one pentad from consecutive training bins only — is rolled from the same nine starts and scored through the same battery (Table 2, Figure 5). At 200 modes (76% of the training variance; spectral radius 0.986, leading e-folding time 360 days) its corridor correlation is 0.41 at five days, 0.26 at 90 and 0.18 at 365; its amplitude falls with lead in step with its correlation, so $\text{MSSS}_{\text{clim}}$ stays positive at every lead (+0.17 at five days, mean +0.054). The head’s correlation exceeds the LIM’s at five and ten days only; from fifteen days on the LIM is more correlated with the truth at every lead, and its mean squared-error score is above both the head’s measured value and the head’s amplitude-calibrated value (+0.019). Against damped persistence the LIM loses at leads up to 15 days and wins by 0.01–0.07 from 30 days out; the head loses at every lead. Fifty and 100 modes score below 200 at every lead (mean correlation 0.200, 0.209, 0.225). Per channel, the LIM holds sea-surface temperature at 0.48, 0.41, 0.33 and 0.28 at 5, 30, 90 and 365 days against the head’s 0.76, 0.34, 0.34 and 0.10, and sea-surface height at 0.57, 0.36, 0.30, 0.30 against 0.84, 0.25, 0.11, −0.06. The LIM’s training set is every bin outside a held-out year (2,923 bins, 2,919 pairs), which is slightly larger

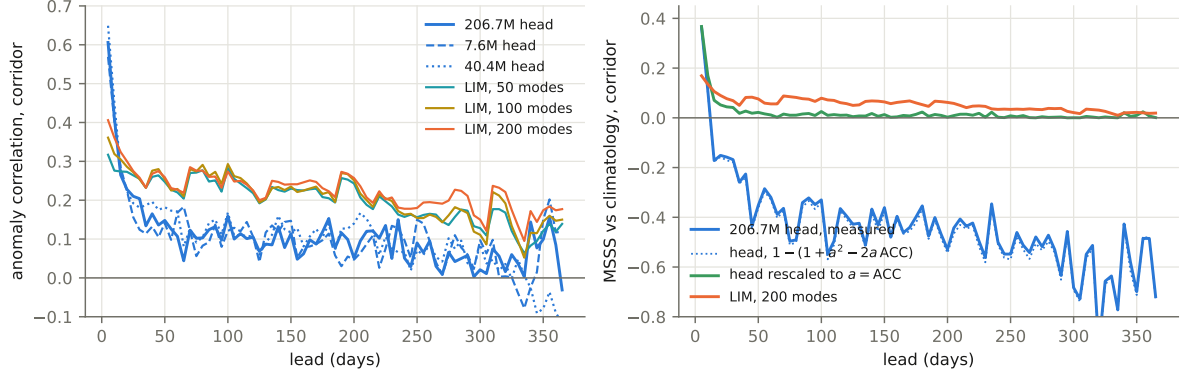


Figure 5: Left: corridor anomaly correlation per lead for the 206.659M, 40.388M and 7.598M heads and linear inverse models with 50, 100 and 200 modes, from the same starts. Right: the head’s measured $\text{MSSS}_{\text{clim}}$, the value the identity predicts from its own ACC and a , the value the same states would score at $a = \text{ACC}$, and the 200-mode LIM.

lead (d)	ACC			$\text{MSSS}_{\text{clim}}$			a		
	7.6M	40.4M	206.7M	7.6M	40.4M	206.7M	7.6M	40.4M	206.7M
5	0.569	0.649	0.606	+0.314	+0.422	+0.365	0.69	0.69	0.69
10	0.403	0.452	0.412	+0.125	+0.177	+0.109	0.61	0.63	0.67
15	0.284	0.276	0.265	−0.005	−0.098	−0.172	0.58	0.69	0.76
30	0.124	0.181	0.204	−0.160	−0.123	−0.168	0.55	0.58	0.67
90	0.137	0.150	0.132	−0.120	−0.189	−0.322	0.52	0.62	0.72
180	0.114	0.140	0.153	−0.139	−0.180	−0.354	0.51	0.59	0.78
365	0.119	−0.116	−0.031	−0.181	−0.593	−0.719	0.54	0.66	0.82
mean, 73 leads	0.103	0.104	0.105	−0.179	−0.257	−0.439	0.54	0.63	0.78

Table 3: The 7.598M, 40.388M and 206.659M heads through the same battery, corridor scope. Mean $\text{MSSS}_{\text{damped}}$ -0.200 , -0.277 and -0.461 ; gate/window mean $\text{MSSS}_{\text{clim}}$ $-0.201/-0.213$, $-0.208/-0.227$ and $-0.395/-0.401$. Single seed each.

than the pool the head’s 144-bin windows admit.

Width on the rolled field. The 7.598M (256×8) and 40.388M (512×12) heads at their 20,000-step end states, rolled through the identical battery, score the same field as the 206.659M head (Table 3, Figure 5): mean corridor correlation 0.103, 0.104 and 0.105 across a $27 \times$ span in parameters, and amplitude-calibrated ceilings (mean ACC^2) of 0.018, 0.022 and 0.019. The three differ in amplitude, which is monotone in width ($a = 0.54, 0.63, 0.78$) and is the whole of the spread in $\text{MSSS}_{\text{clim}}$ ($-0.179, -0.257, -0.439$); the identity above reproduces each head’s 73 leads to a mean absolute error of 0.006, 0.007 and 0.014. They also differ at the ends of the lead axis. The 40.388M head is the most correlated over the first two pentads (0.649 and 0.452) and is the only head with a positive $\text{MSSS}_{\text{damped}}$ at any lead (+0.08 at five days); the 7.598M head does not decay through zero, holding 0.10–0.14 from 60 days to a year where the two larger heads fall to -0.12 and -0.03 , and its a does not grow with lead. Sea-surface temperature carries the skill in all three (mean $\text{MSSS}_{\text{clim}}$ $-0.006, -0.143, -0.165$; positive at 33, 15 and 11 of 73 leads); sea-surface height is the sharpest one-step channel for all three (0.87, 0.87, 0.84). All three beat raw persistence at 71 or more of 73 leads, and the linear inverse model is more correlated than any of them from fifteen days on. The transport bands of the three heads disagree in sign ($-0.025, +0.103, -0.244$; $-0.099, -0.131, -0.310$; $+0.107, -0.242, +0.163$). Roll times 7,660, 21,077 and 77,147 s. One seed each.

Per channel. Only eight of the forty channels are scoreable. Sea-surface temperature holds correlation longest (0.76 at five days, 0.34 at 30 and at 90) and is the only channel whose $\text{MSSS}_{\text{clim}}$ stays positive beyond a few pentads; sea-surface height is sharpest at one step (0.84) and below 0.2 by a month; current speed and mixed-layer depth are one-step quantities; the four wind channels are noise past one

channel	ACC 5 d	ACC 30 d	ACC 90 d	mean MSSS _{clim}	mean MSSS _{damped}
sst	0.759	0.343	0.337	−0.165	−0.191
ssh	0.838	0.252	0.113	−0.499	−0.562
cur_speed	0.604	0.131	0.052	−0.591	−0.606
log_mld	0.572	0.178	−0.009	−0.464	−0.469
tau_x	0.490	0.097	0.022	−0.383	−0.386
tau_y	0.235	0.039	−0.017	−0.426	−0.428
tau_x_std	0.222	0.134	0.104	−0.409	−0.410
tau_y_std	0.207	0.070	0.057	−0.465	−0.466

Table 4: Per-channel corridor scores of the rolled head.

step (Table 4).

Transport. The unpooled read-out of the rolled section states against RAPID gives $r = +0.107$ over 5–90 days ($n=162$), -0.242 over 95–180 ($n=129$), $+0.163$ over 185–365 ($n=150$). With about nine effective independent starts these are inside the noise of zero. A separate 36-month free run from a context ending 2004-12 — inside the training years, so not a skill measurement — correlates 0.636 with the true transport at pentad resolution and -0.13 on an 18-month low-pass at 61% amplitude: what the head tracks lives at high frequency, and the multi-year band is not measured by this roll.

Statistical power. The pool has 2,417 distinct end-times; consecutive windows share 143 of 144 frames, and every pixel is a correlated view of the same temporal patterns. With 206M parameters that is $\sim 85,000$ parameters per pattern, and the training loss falling long after the held-out loss has turned is the expected consequence. For field-level scores the effective sample count is in the thousands and differences can be resolved; for the transport target (RAPID ~ 20 years, anomalies decorrelating over months to years) it is of order ten, and a band correlation of ± 0.2 cannot be. Every rolled number here is one seed; rolled skill at this cadence has not been replicated, and no two rolled numbers can yet be called different.

4 A cone-based design for the next codec and forecaster

The measurements above fix what a redesign has to answer. The codec is at parity with the raw fields it compresses for a current-state read-out, so its only value is as a substrate for a forecaster (§2); the forecaster learns what transfers inside two thousand steps at every width, memorises afterwards, emits anomalies whose amplitude does not fall with its correlation, loses to a damped-persistence null at every lead and to a 200-mode linear inverse model from fifteen days on (§3); and the field, not the transport index, is the only instrument that can resolve a design choice (nine effective starts at 26.5°N). This section specifies both stages from those constraints. Nothing in it has been trained; every number that is a choice is marked as one, and §4.4 names the comparison that would settle it.

Both stages read the same shape. A pixel’s future depends on a cone in space–time: a driver moving at speed v reaches the anchor from within a distance $v \Delta t$ after Δt , so the set of space–time points that can bear on the anchor widens into the past, and, by the same argument run forward, the set the anchor can bear on widens into the future. Both stages sample that double cone — an hourglass with the present at its waist — and differ only in what fills it and at what scale (Figure 6). Stage 1 reads raw values over the last 30 days and writes one embedding per cell and pentad; stage 2 reads those embeddings over the last two years and writes the embeddings of the next year. In both, the past half of the hourglass is input and never a forecast target, the waist is the present, and the future half is target only.

4.1 Data and tokens

Grid. The global 0.25° pentad tensor ($721 \times 1,440$ cells, 1982–2024, 3,142 bins) in its four native-resolution groups: 7 ocean channels at 0.25° (current speed and its two components, sea-surface height,

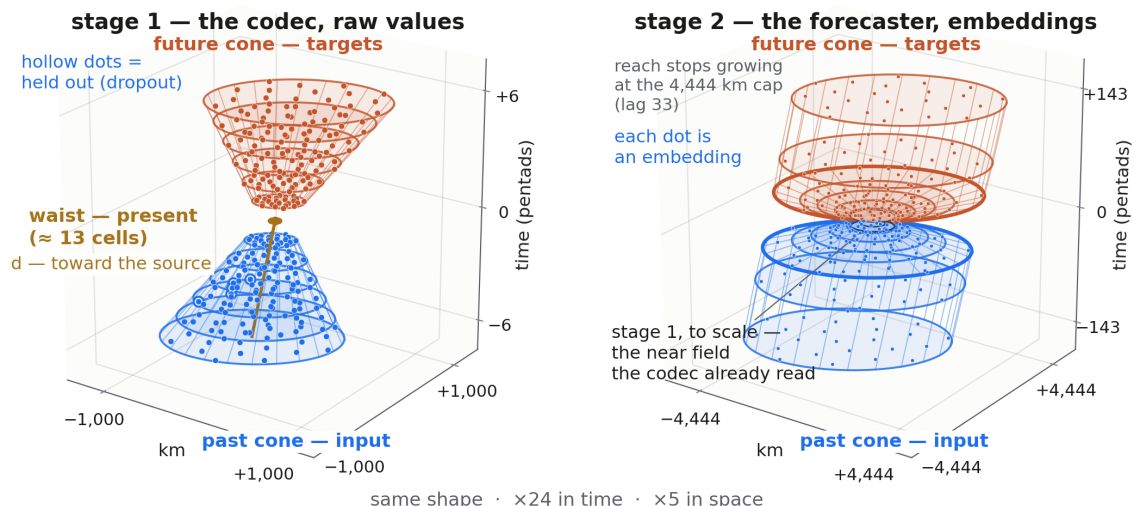


Figure 6: The hourglass at both scales. Each lag is a footprint of 24 sampling points (a Vogel sunflower, phase-rotated by the golden angle per lag so consecutive slices interleave) inside an ellipse whose size grows with lag. Left: stage 1 over raw values, six pentads each way, ≈ 900 km at six pentads for the ocean-surface channels, a waist of thirteen cells. Right: stage 2 over stage-1 embeddings, 7–143 pentads each way, opening to a 4,444 km cap; stage 1’s whole hourglass is the small solid at the centre, the near field the codec has already read.

sea-surface temperature, sea-ice concentration, log mixed-layer depth), surface chlorophyll-a at 0.25° from late 1997, 15 atmosphere-and-land channels at 1° (wind stress and its within-pentad variability, 2 m air and skin temperature, 10 m wind, surface pressure, precipitation, snow water equivalent, soil moisture and temperature, the two turbulent heat fluxes), and 32 ocean-interior channels at 1° and monthly cadence (temperature and salinity at 16 pressure levels), with two statics (surface class — ocean, land, ice sheet, inland water — and elevation). A coarse channel is read at its own cell, so nine fine cells under one 1° cell see one number; that is the model’s honest view and is not interpolated. No channel is masked by sphere: a land cell reads its skin and air temperature and soil state, an ocean cell its surface fields, and every cell reads the shared atmosphere.

Anomalies and scaling. Every channel enters as a standardised anomaly against a day-of-year harmonic climatology (annual and semi-annual terms) fitted on training years only, divided by a training-year standard deviation per channel; nothing in the climatology, the scale or the normalisation of any read-out sees a held-out year. Log-normal quantities (mixed-layer depth, chlorophyll, precipitation, snow water equivalent) are transformed before standardisation; bounded fractions (sea-ice concentration, soil moisture) through a logit. A value that does not exist at that cell and bin — a land cell’s sea-surface temperature, a month without an Argo profile — is not a zero: each token carries an *observed* flag per channel, and an unobserved entry is masked from the loss and from the encoder’s input.

Per-location tokens. The unit the encoder attends over is a place at a time: one token per sampled cell and lag carrying the value vector of every channel present there, the observed-flag vector, the offset $(\Delta x, \Delta y, \Delta t)$ from the anchor in kilometres and pentads, a sphere code, the local day-of-year harmonics and the latitude. One token per dot, not one per channel per dot: the channel dimension lives inside the token, so the token count is the geometry’s and not the channel count’s, and adding a channel costs a vector entry rather than a token.

Point observations. Measurements that are not on the grid — Argo profiles, surface drifters, moored buoys, along-track altimetry, surface-ocean carbon samples — enter as tokens of the same schema placed where and when they were observed: value, offset from the anchor, a footprint (the measurement’s own spatial and temporal extent), instrument and channel identifiers. The k nearest profiles in space-time within the cone are read, k small (three to eight per depth group); there is no ellipse to learn for them, only the attention over what was measured. Datasets whose licences forbid redistribution are read on

the same footing and never republished.

4.2 Stage 1: the hourglass codec

Geometry. The present is a *waist* of about thirteen cells (a disc of radius two cells at 0.25°), read and reconstructed, and the embedding z is written for the anchor at its centre. Gradients need neighbours: thermal wind and geostrophy are differences across cells, and a single-cell present cannot hold one. The past cone samples lags 1–6 pentads, every pentad, so the codec owns the last 30 days at full cadence. At each lag and for each channel group the dense sampling set is a 24-point sunflower mapped through an ellipse of covariance $\Sigma(\ell) = \Sigma_0 + \ell^2 \Sigma_v$ centred at $\ell \mathbf{d}$ from the anchor; initial reaches follow the fastest measured signal of the group (130 km per pentad for the ocean surface, ≈ 900 km at six pentads; 500 km for wind stress within its ten-day memory; the sea-surface temperature and mixed-layer channels wide at lags 0–1 where the atmosphere stirs them and then the ocean’s growth; 400 km flat for land, which does not advect), and a sparse fixed far ring extends every family to the reach of the fastest signal of its kind at $1.5\times$ a design speed of 5.4 m/s, so that a driver is never excluded by construction. The *prediction cone* is the past cone point-mirrored through the anchor, leads +1 to +6 pentads, sampled the same way and used only as targets; the anchor column is always among them. Each channel reads the group’s shared tokens through its own Gaussian aperture $w_{c,i,\ell} = \exp(-\frac{1}{2} p_i^\top \Sigma_c(\ell)^{-1} p_i) \exp(-\ell \Delta t / \tau_c)$ applied to that channel’s entries before the projection — one scale s_c and one memory τ_c per channel, one direction per flow — at zero extra tokens.

Learnable geometry. Eight dimensionless numbers per 1° cell and channel group are learned: the drift $\mathbf{d} = \tanh(\mathbf{x}) v_{\text{design}} \Delta t$, capped at the design speed by construction, and the two ellipses stored as matrix logarithms, $S = \log \Sigma = m\mathbf{I} + k [\cos 2\theta, \sin 2\theta; \sin 2\theta, -\cos 2\theta]$, so that no parameter is in kilometres, no angle wraps, and an Adam step of one learning rate moves every radius by the same fraction of itself whatever its size. The maps are stored as a coarse 10° field plus a 1° residual under an L2 and Laplacian prior, so data-poor cells inherit their neighbourhood. The geometry learns in two phases: first through the apertures alone over a fixed candidate set (a log-radial dot set out to the design reach), so the data loader’s reads are regular and never depend on live weights and the gradient on $\Sigma_0, \Sigma_v, \mathbf{d}, s_c, \tau_c$ is informed by every candidate at once; then the dense dot table is re-baked from the learned ellipses every 2,000 steps and swapped at an epoch boundary. The far ring is never gated, target positions carry a stop-gradient, the future cone is tied to the past cone by the mirror, the evaluation geometry is frozen for every arm, and the aperture never enters the loss weights: a learnable cone can otherwise learn to ask easy questions, and each of those rules closes one way of doing so.

Tasks. Each training example is one anchor at one bin and draws one of four tasks: forecast the prediction cone from the waist and the past cone (share 0.35); forecast it from the waist alone with the past fully hidden (0.15), the snapshot twin inside one model, so every run reports what history buys; nowcast the hidden waist from the past under a dropout pattern (0.25); and fill in hidden waist cells and hidden past dots from what remains (0.25). Dropout patterns fire independently: one channel hidden at the waist, the most recent lags hidden so the model must extrapolate from stale history, a 90° bearing wedge hidden so it cannot lean on one direction, dots hidden at random to imitate sparse observation, and, for the nowcast only, the whole waist. The one rule that is not a knob: a target’s channel must be visible somewhere in the input, at some lag, or the target is unknowable and teaches the model to predict the mean; the sampler enforces this and the log counts violations (required zero). Shares are recorded per run and are the first thing the comparison in §4.4 varies.

Model. A masked autoencoder in the strict sense: the encoder sees only visible tokens and visible channel entries, never a placeholder for a hidden one. The encoder is a transformer over the ≈ 300 per-location tokens of one example (Perceiver-style cross-attention into a fixed latent set when a fixed-size interface is wanted; a plain encoder with a learned attention pool otherwise), producing z of width 128–256 — wider than the 32 used above, because the compression the forecaster needs is in the token count, not in the width, and a narrow bottleneck was where capacity fights between the waist and the dots showed. The decoder is query-based and reads z alone: a query is (offset, channel, sphere, depth, lead), it cross-attends to z , and its head emits a mean and a log-variance per channel group. Because the query carries the channel and the offset, “channel c at point p at lead ℓ ” is one query, and the same

decoder answers a grid cell, a station and a float profile. A decoder that could also read the encoder’s latents would let z carry nothing; that path is closed.

Objective. A heteroscedastic Gaussian likelihood per channel group in its transformed space, in the decoupled form

$$\mathcal{L} = \|\mu - y\|^2 + \frac{1}{2} e^{-s} (y - \text{sg}[\mu])^2 + \frac{1}{2} s, \quad s = \log \sigma^2,$$

so the mean is trained by plain squared error — what the forecaster consumes and what every score here measures — while the variance is trained by the likelihood with the mean detached, which removes the pathology in which the variance head explains a hard target away instead of the mean learning it. s is clamped to $[-8, 8]$, the first 2,000 steps run at fixed $\sigma = 1$, and every family — waist reconstruction, dot fill-in, each future lead — is scored against its own predict-the-mean bar, logged at every evaluation; the bar is what says which family carries signal, and a family whose held-out error does not fall below it is reported as carrying none. Channel groups are weighted explicitly in the loss as a stated choice, never through σ as a side effect.

Training stability. The failure modes that have been measured each get a guard. Embedding collapse is invisible to reconstruction loss (a decoder rescales freely), so the effective rank of z over a fixed held-out batch and a scale-invariant probe correlation are logged at every evaluation and the run aborts on two consecutive sub-threshold readings; no quantisation is switched on from a random initialisation, since a lattice on an untrained encoder collapses within thousands of steps and a lattice added to a finished continuous codec does not. Inputs are standardised, so the gradient clip is 1.0; AdamW with $\beta_2 = 0.95$, weight decay 0.1, a 2,000-step warm-up and cosine decay to 10% of peak; bf16 with fp32 master weights; the geometry parameters get a learning-rate multiplier of ten (a choice, recorded per run). Training stops at the held-out minimum with the minimum’s step recorded, and no arm runs past four epochs of anchors. Held-out years are excluded from every window the forward pass can see — every past-cone dot, every future-cone target — and the exclusion is certified by brute force before training.

4.3 Stage 2: the forecaster

Inputs. Stage-2 tokens are the codec’s embeddings: at each sampled cell and lag, the mean μ and log-variance s of z , the offset $(\Delta x, \Delta y, \Delta t)$, the statics and the day-of-year harmonics of that bin. The past cone reads lag 0 (which already summarises the last 30 days) and log-spaced lags 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 pentads — twelve slices instead of 144 — each with the anchor column and a 24-dot footprint whose reach follows the fastest slow signal (planetary waves and gyre adjustment at 0.05–0.1 m/s: $\approx 1,000$ km at 34 pentads, the 4,444 km cap from about lag 33 on). Dense sampling where predictability is dense and sparse where it is not is what keeps the example at ≈ 300 tokens rather than 21,000; the dense 144-pentad window is the ablation. The same eight-number learnable geometry applies at this scale, with its own 1° maps and the same guards.

The linear backbone. The linear inverse model out-forecasts the learned head from fifteen days on and a damped-persistence null beats it at every lead. The design therefore makes the learned model a correction to a linear one rather than a replacement for it. For the anchor column,

$$\hat{z}_{t+\ell} = \mathbf{G}_\ell z_t + f_\theta(\text{cone}, \ell), \quad \mathbf{G}_\ell = \mathbf{U} \text{diag}(e^{-\ell/\tau_j}) \mathbf{U}^{-1},$$

where $\mathbf{G}_\ell$ is a linear propagator in embedding space with learned time constants $\tau_j > 0$ — contractive by construction, so a free run decays toward climatology rather than drifting — initialised from a linear inverse model fitted to the training-year embeddings, and f_θ is the transformer over the cone with a zero-initialised output projection. At step zero the forecaster *is* the embedding-space linear inverse model; whatever it becomes has to earn its distance from it on the held-out field. Damped persistence is the diagonal special case and is reported as its own null.

Direct multi-lead targets. The target is the mirrored future cone: the anchor column and its footprint at leads 1, 2, 3, 5, 8, 13, 21, 34, 55, 73 pentads (73 pentads is one year), each lead a query $(\Delta x, \Delta y, \ell)$ to the same decoder head, predicted directly rather than by chaining one-step predictions. A direct

forecast at lead ℓ cannot compound its own errors, its amplitude is set by the data at that lead rather than by an accumulation, and the log spacing puts the targets where the correlation is still changing. The anchor column is weighted 1 and the off-anchor footprint 0.25 (a choice); the footprint targets are what make the prediction a field with coherent advection rather than a set of independent columns. Leads beyond a year are reached by feeding the predicted anchor embeddings at leads 1–73 back as the recent past and querying again — the multi-year roll is a second mode of the same model, not its training objective.

Objective and calibration. The decoupled Gaussian likelihood on each target embedding, normalised per lead by that lead’s predict-the-mean bar so that every lead counts equally and the long, unpredictable leads do not dominate the gradient; plus a decoded-space term through the frozen codec decoder for the scoreable channels at leads 1, 8 and 34 (squared error against the standardised anomaly, relative to climatology), because the embedding metric is not the metric the forecast is scored on; plus, for the multi-year mode, a two-step pushforward term in which the model is rolled once with a stop-gradient on the first step and scored on the second, so it learns to correct its own errors without back-propagating through the roll. Under a squared-error mean the conditional expectation shrinks toward zero as correlation falls, so amplitude calibration is a property of the objective at the held-out minimum and not a post-hoc fit; the anomaly-correlation-amplitude identity of §3 is logged per lead as the check that it holds, and a per-lead scaling fitted on training-year starts is reported as a diagnostic only.

Regularisation and memorisation. The held-out minimum inside two thousand steps and the $30\times$ train/held-out gap say the previous forecaster had $\sim 85,000$ parameters per distinct temporal pattern. Three things change that. The globe gives every regime of the ocean and atmosphere as training patterns rather than one basin’s; the log-spaced cone reduces the parameters per example by a factor of seventy; and the model’s free part is a residual on a linear backbone that the data pin down with far fewer degrees of freedom. On top of these: Gaussian noise on the input embeddings at 0.7 of the persistence scale, as in §3; random dropout of past-cone dots; weight decay; early stopping at the held-out minimum with the step recorded; and, because the model is cheap at this token count, a seed ensemble of three to five whose spread is the first probabilistic forecast and whose mean is the deterministic one.

Probabilistic extension. Once the deterministic level exists, a shared low-dimensional noise vector per forecast conditions the head so that samples are coherent across cells and leads; scored by the fair continuous ranked probability score. Explicit categorical marginals over a quantised embedding are the alternative where the conditional distribution is two-humped, which at six to twenty-four months the El Niño index plausibly is; both are decided on the dense field, never on an index.

Read-outs. Every quantity of interest is a head on the predicted embeddings: transport at 26.5°N through the unpooled attention head over the section’s cells (§2), trained on training-year true embeddings and applied to forecasts; the Niño-3.4 index through the same head shape over its box; any channel at any cell through the codec decoder. The read-out is the cheap, replaceable end of the system, and none of them is used to choose between models.

4.4 Evaluation and the comparisons that decide the design

Protocol. Held-out years 2009, 2017 and 2023 stay held out of training, of the climatology and of every statistic; a codec and forecaster trained on ≤ 2020 are scored once on 2021–2024 as a terminal test. The battery is the one above: for each lead, mean-squared skill against climatology, raw persistence and damped persistence, anomaly correlation and amplitude ratio, per channel and per scope, from three starts per held-out year, with block-bootstrap intervals over starts. The null ladder beside every read-out is persistence, damped persistence, the linear inverse model in data space, the linear inverse model in embedding space, and an analogue (retrieval) forecast; a learned model is reported against all five. A configuration is compared to another on the dense field; the transport and index read-outs are downstream checks and cannot resolve a difference at nine effective starts. The winning configuration of any comparison is replicated with a second seed before it is written up.

The comparisons, in order. (1) The hourglass codec against the current codec, both frozen, both feeding the same forecaster: the one-step forecast ratio and the future-cone families against their bars. (2) The learnable geometry: a fixed hourglass, a learned one initialised at the fixed one, and a learned one whose surface drift is initialised at the reversed climatological current — three seeds each, because the read-outs include a geometric one (at the 26.5°N array the learned surface drift should point south-west up the Florida Current, and attention on deep profiles north, up the deep western boundary current); the learned arm is adopted if it beats the fixed arm on the future-cone loss at every lead ≤ 3 , paired at all seeds, with the target-variance flag clean, and if only the primed arm wins the geometry is set from climatology rather than learned. (3) The forecaster: the linear backbone with a residual against the same transformer without it; direct multi-lead targets against one-step autoregression rolled; log-spaced against dense lags; each on the same codec, the same starts and the same battery. (4) The far ring: the learned codec without it, which measures whether the ring pays for its tokens rather than arguing it. Each comparison is one line in a configuration; none affects the build order, which is geometry, task mix, aperture gate, parameter maps and harden step, forecaster.

5 Related work

Observing the overturning. Continuous measurement of the Atlantic meridional overturning is two decades old. The RAPID-MOCHA-WBTS array at 26.5°N has measured the transport since April 2004 as the sum of the Florida Current cable, the Ekman transport from wind stress and the geostrophic interior from boundary density moorings [1]; the record reads $17.0 \pm 2.8\text{ Sv}$ with a trend of -1.0 $[-0.4, -1.6]$ Sv per decade that is marginally significant and dominated by a 2004–2010 drop and a 2010s pause [2, 3], with the climate signal carried by the deep and thermocline components rather than by the Gulf Stream or Ekman terms [2]. The Florida Current itself shows four decades of steady state once the cable is corrected for geomagnetic secular variation [4]. MOVE at 16°N monitors the deep western boundary since 2000 [5]; OSNAP measures the subpolar overturning since 2014 and places most of it east of Greenland [6, 7]; a 2026 synthesis of four western-boundary arrays reads a meridionally consistent decline in deep western transport between 16.5° and 42.5°N , partly compensated at the eastern boundary [8]. Whether the overturning weakened before 2004 is contested between sea-surface-temperature fingerprints [9], which read a $\sim 15\%$ decline, and air–sea heat-flux fingerprints, which read none since the 1960s [10]. Our transport target is the RAPID series deseasonalised at monthly and pentad resolution, with MOVE as a second target, and our inputs carry no transport measurement.

Reconstructing transport from observables. The classical line regresses the 26.5°N transport on physically chosen observables: satellite sea level with the cable and Ekman terms [11], sea level through a thermal-wind argument [12], or boundary densities [13]; these reach r of 0.8–0.95 on 18-month low-passed series with in-sample calibration, and the one out-of-sample case holds out a single year. The machine-learning line trains on simulations: a convolutional network reading bottom pressure, sea level, temperature and wind strips reaches $r \approx 0.98$ inside a laminar 1° state estimate [14], the same task in eddy-resolving models explains 0.44–0.74 of the variance [15, 16], and a graph network on Argo profiles, wind stress and the boundary currents trained on VIKING20X recovers about 80% of the seasonal variance at 26.5°N with monthly errors under 1 Sv [16]; deep-learning reconstructions from surface fields trained on climate models extend an index back to 1900 [17]. Our read-out differs from all of these in protocol rather than in method: real RAPID data, monthly or pentad resolution with no low-pass filter, no transport among the inputs, and a year-blocked k-fold in which every year is held out in turn. On those terms a frozen embedding reads transport at $r = 0.67$ and a raw-pixel head at 0.66 (§2), which is where the eddy-model results place the difficulty of the task; the same read-outs low-passed at 18 months score 0.33–0.75 across codec seeds on nine effective points, which is why the report quotes unfiltered numbers.

Forecasting the overturning. Initialised decadal prediction systems show positive skill for subpolar ocean heat content at lead years 2–10 and for coastal sea level along the North American east coast at 3–5 years [18, 19], but the direct transport record is too short for the overturning itself to be verified at those leads; at 26.5°N the signal-to-noise ratio of the trend is not expected to exceed 2 before about 2040 [2]. The verification framework used here — mean-squared skill scores against climatology and persistence, anomaly correlation per lead, a damped-persistence reference — is the one established for interannual-

to-decadal prediction [20]. The linear inverse model [21] is the standard empirical forecast against which such systems are judged, including as a benchmark for decadal forecasts of surface temperature [22]; the version in §3 fits it in the same standardised field the transformer is scored on. Statistical early-warning analyses of sea-surface-temperature fingerprints have placed a tipping of the overturning in the mid-21st century [23], but the estimate moves by a century or more with the choice of dataset and fingerprint [24], and physics-based indicators such as the freshwater transport at 34°S [25] are the alternative; none of this is a forecast at the pentad-to-seasonal leads measured here, where the question is whether any learned model adds to a linear operator.

Learned ocean and climate forecasting. Neural ocean forecasters at 1/4° to 1/12° and the global weather models they descend from [26, 27] operate at daily leads on full reanalysis fields and are cited for orientation; at monthly cadence, convolutional ENSO forecasts hold Niño3.4 correlation above 0.5 to about 17 months [28] and sea-surface-temperature anomaly fields are forecast to 18 months with recurrent networks [29]. Architecturally the codec is a masked autoencoder [30] over channel tokens, and the frozen-representation evaluation used throughout follows the argument that frozen backbones with shallow read-outs are the honest measure of a foundation model’s representation [31]; the closest statement of the per-pixel embedding-field idea is terrestrial [32].

6 What the results support

For the current-state transport read-out the codec is at parity with the raw fields it compresses; its value, if any, is as a substrate that a transformer can attend over and roll forward, and on that use the transformer has learned what it learns inside two thousand steps at any width, rolls the same field at 7.6M, 40.4M and 206.7M parameters, forecasts sea-surface temperature and height for days to weeks, does not beat a damped-persistence null at any lead, and is out-forecast beyond ten days by a linear inverse model with 200 modes fitted to the same field. The linear model is therefore the reference against which any further change to the forecaster is scored: what a learned model adds at leads of one to two pentads, and whether any change beats the linear model at three. The next measurements are the ones that bound this further: a retrieval baseline and a linear inverse model in embedding space through the identical battery, with block-bootstrap intervals; the early-stopped checkpoints rolled against the end states shown here; amplitude calibration fitted on training-year starts; and a codec trained on ≤ 2020 so that a terminal 2021–2024 test can be run once. §4 is the design these measurements motivate — the same hourglass at both stages, a codec that owns the last thirty days of raw values, and a forecaster that is a residual on the linear model rather than its replacement — and §4.4 the comparisons, in order, that would accept or reject each of its choices.

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A Data and code availability

Code: <https://github.com/blauewelt/earth> (codec trainer `ml/train.py`, stage-2 trainer `ml/temporal.py` and its JAX port `ml/jaxport/`, evaluator `ml/rollout_spatial.py`). Result bundles are on the repository’s `ml-metrics` branch (`probes-<n>.json`; the rolls here are `probes-516.json`, `probes-520.json` and `probes-523.json`, the linear inverse model `probes-527.json` (`ml/lim_baseline.py`), the codec probe ladder is in the bundles named in `ml/EXPERIMENTS.md`). Weights are on the `model-checkpoints-v1` release (`run-415__pixelmae.pt`; `head-weights-e059-200k-window-s0.pt` and the `e060a/e060b` heads named likewise); the tensor on the `data-cache-v1` release and the Hugging Face dataset `chfrank/earth-tensors`; the codec’s embedding of it on `embed-cache-v1`. Stage-2 figures and tables are generated by `ml/paper/make_figs.py` from `ml/paper/data/report_data.json`, built by `ml/paper/extract_data.py` from those bundles and the training records; codec probe numbers are transcribed from the archived per-run bundles, with the source named beside each in the figure script.

Sources: GLORYS12 and the GREP ensemble (E.U. Copernicus Marine Service Information, used under the CMEMS licence with credit to the service); the Roemmich–Gilson Argo climatology (Scripps; Argo data are collected and made freely available by the International Argo Program and the national programs that contribute to it); NCEP/NCAR Reanalysis-1 (NOAA PSL); OISST v2.1 (NOAA NCEI); the RAPID-MOCHA-WBTS transport time series (NERC, NSF, NOAA); the MOVE transport time series (NOAA).